Distribution-Preserving QIM

Histogram detectability and robustness trade-offs

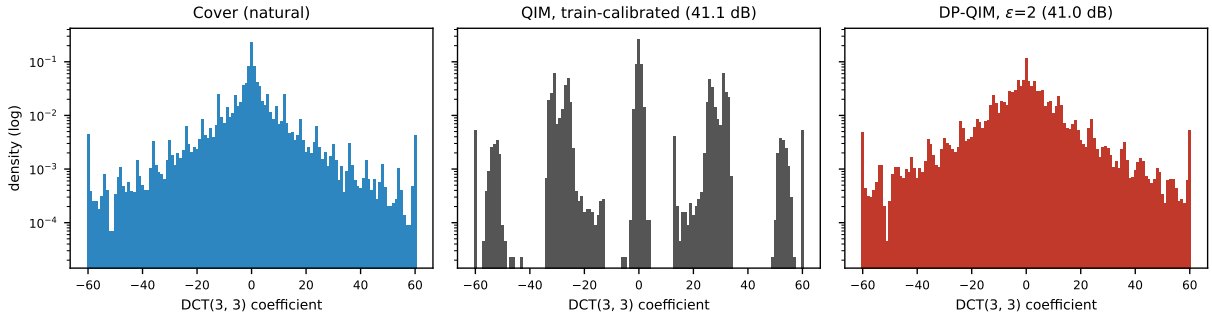
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https://github.com/sairam782/Invisible-Watermarking

Corrected result, not a security guarantee

DP-QIM replaces a mismatched DCT carrier with a sample from the target quantile cell instead of a lattice point. It reduces some histogram fingerprints but remains detectable. The previous near-undetectability claim is withdrawn.

What the theory actually says

For ideal continuous equal-mass cells and independent uniform bits, $P_W = \frac{1}{2}F + \frac{1}{4}G_{0,\epsilon} + \frac{1}{4}G_{1,\epsilon}$, where each $G_{i,\epsilon}$ is conditioned *within its own safe cell*. Global conditioning is not equivalent in general. Finite pools, ties and RGB rounding prevent an exact implementation-level guarantee. A zero decision boundary is invariant to positive scalar scaling; this is not a theorem about image blur or resizing. QIM can decode correctly after large displacements because its label regions repeat periodically.



Protocol and corrected measurements

43 crops from 14 distinct photos; exact source/crop hashes checked. The `cat`/`chelsea` alias is removed. Every detector component and QIM fidelity calibration uses the same outer training split. AUC is averaged within folds, not pooled across different classifiers. Intervals are conditional source bootstraps, not uncertainty over retraining.

Detectability (0.5 = chance)

	Learned detector		Peak statistic	
Bits (PSNR)	DCT-QIM	DP-QIM	DCT-QIM	DP-QIM
64 (53.4)	0.95 [0.91, 1.00]	**0.65 [0.63, 0.76]**	0.64 [0.52, 0.77]	**0.47 [0.41, 0.49]**
256 (46.9)	1.00 [1.00, 1.00]	**0.84 [0.79, 0.92]**	0.83 [0.67, 1.00]	**0.50 [0.39, 0.59]**
1024 (41.0)	1.00 [1.00, 1.00]	**0.96 [0.91, 1.00]**	0.97 [0.92, 1.00]	**0.62 [0.40, 0.80]**

Robustness (matched-PSNR difference)

Attack	DP-QIM	Δ vs. DCT-QIM (pp)	95% CI
No attack	0.999	-0.1	[-0.2, +0.0]
JPEG Q=90	0.834	**-15.8**	[-19.8, -12.7]
JPEG Q=70	0.704	**-28.6**	[-32.7, -25.3]
Gaussian noise σ=0.03	0.830	**-14.1**	[-17.0, -11.5]
Gaussian blur r=1	0.841	**+34.8**	[+30.2, +40.2]
Downscale 0.5×	0.826	**+33.4**	[+30.0, +36.8]
Contrast ×0.5	0.967	**+20.8**	[+12.8, +32.4]
Contrast ×0.8	0.976	-0.4	[-2.6, +2.3]
Histogram jitter (τ=2, j=0.5)	0.973	**-2.7**	[-5.4, -0.9]
Histogram jitter (τ=1.5, j=1)	0.912	**-7.0**	[-10.0, -4.4]

Interpretation and limits

Learned-detector AUC for DP-QIM rises from 0.65 to 0.84 and 0.96 as payload increases. At 64 bits, blur/downscale/half-contrast gains are 34.8/33.4/20.8 pp, while JPEG 90/JPEG 70/noise cost 15.8/28.6/14.1 pp. Histogram-jitter losses are 2.7 to 7.0 pp; this is *not* published UnMarker. The data were used during development. Independent benchmarks, stronger detectors and gain-invariant QIM baselines remain necessary. Read the full paper and `CORRECTIONS.md`; this summary is not publication validation.